Oasis 3D Printer

Operations Report & Standard Operating Procedure (SOP)

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Software: Oasis_printer_v1.2.1.py + Study_params_loader_and_print.py + campaign_runner.py

1. System Overview

1.1 Oasis_printer_v1.2.1.py

A PyQt5-based GUI application that simultaneously controls an HP45 inkjet printhead and a GRBL motion controller, implementing Binder Jetting 3D printing onto ceramic powder layers.

Component	Description
MainWindow	Main GUI. Split into self.ui (separate window from .ui file) and self.form (widget references)
CameraController	Per-layer image capture (separate widget)
GRBL	Motion serial communication (XY + Z/A rollers)
HP45	Inkjet printhead serial communication
ImageConverter	SVG/image → print array conversion

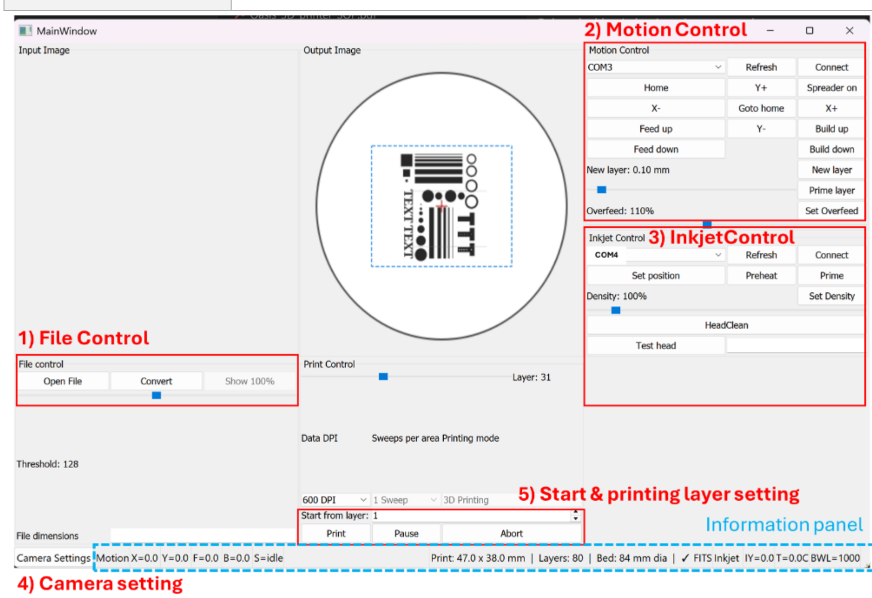


Image for MainWindow.UI

2. SOP — Standard Operating Procedure

2.1 Pre-Run Checklist

Step	Check Item
1	Review parameter values in oasis_change_parameters.csv (especially variance entries)
2	Confirm GRBL motion controller COM port (Device Manager)
3	Confirm HP45 inkjet controller COM port
4	Visually inspect powder fill level and printhead condition

Note : Additionally, the system must **monitor the powder reservoir levels (should be around 70mm below)** to ensure adequate supply and confirm that the **build substrate is cleared** of any previously fabricated structures

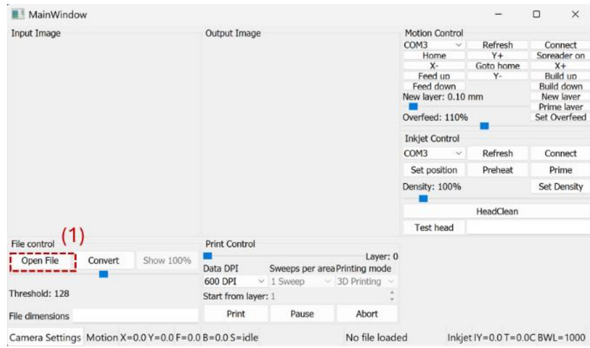
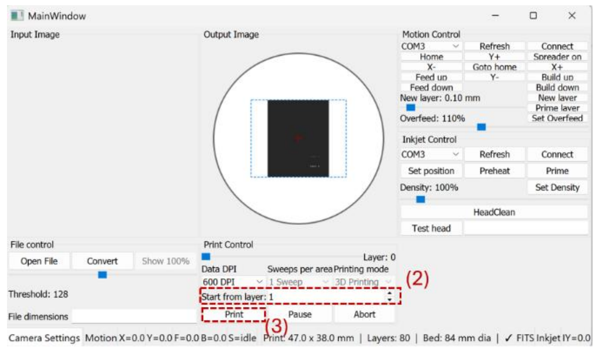
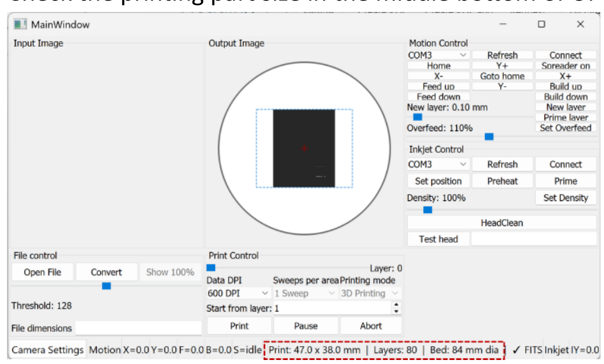
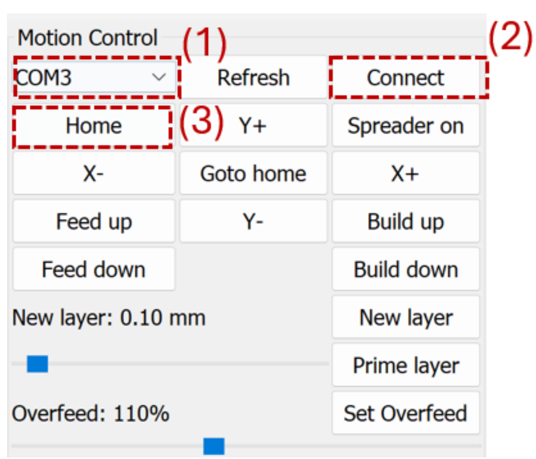
Command : you need to type on the terminal

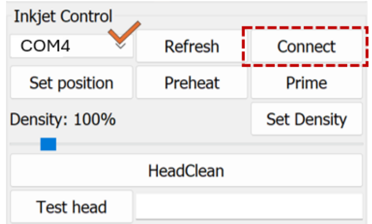
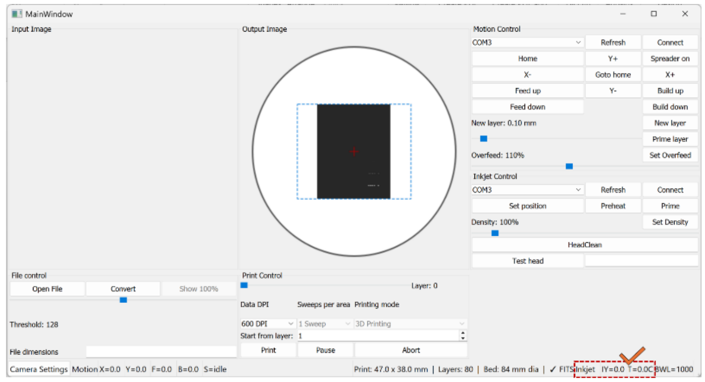
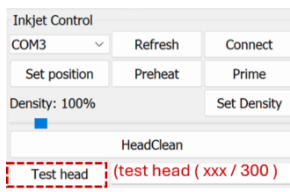
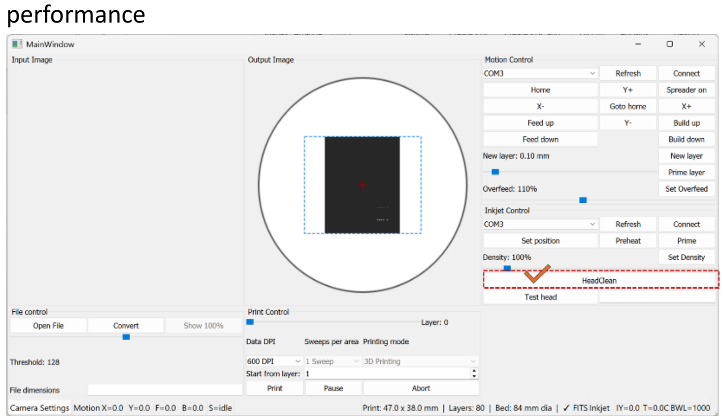
```
- Cd Oasis_3DPrinter
- venv\Script\activate.ps1
- Python Oasis_printer_v1.1.3.py
```

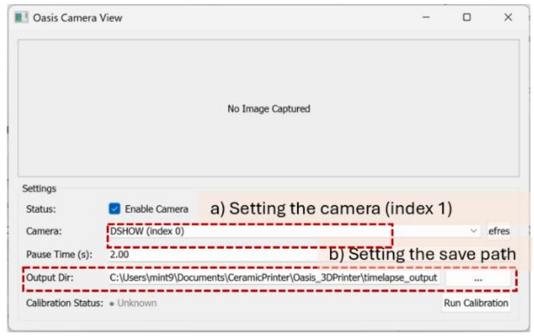
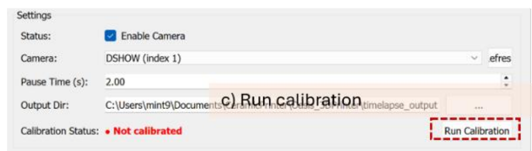
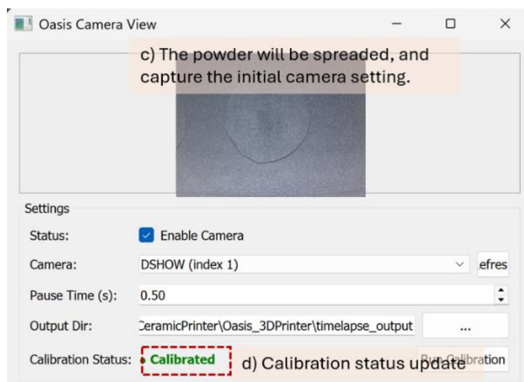
2.2 GUI Startup Sequence

⚠ **The following sequence must be followed.**

Step	Action	Detail
1	**Load SVG File (First Priority)**	File control panel 1. Open File → Select the target SVG file. 2. Need to check the maximum height will be fit in the range (middle of the bottom UI) **Select the svg file you want to print out**

		  Check the printing part size in the middle bottom of UI 
2	Connect Motor COM Port	Motion Control Panel → Select COM port (for Zwick laptop, COM3) → Click Connect 
3	Run Homing	Click Home button — confirm machine returns to origin position before moving the part (there will be error message if you try to move others before click Home button)
4	Connect Printhead COM Port	Inkjet Control Panel

		→ Select COM port (for Zwick laptop, COM4) → Click Connect 
5	**Verify Temperature Reading**	Confirm T (temperature) value appears in the bottom-right of the information panel — indicates successful printhead connection 
6	**Test printhead working**	Click Test head and see 300/300 (meaning all 300 point works okay) 
7	**Printhead cleaning**	**Cleaning Procedure (if poor jetting is observed)** 1) Pink solution sprayed onto fine-cleaner 2) Softly remove clogged binder at the surface of the prinhead 3) Reinstall the printhead 4) Click "HeadClean" in UI and wait for 40s 5) See the binder is jetting when click "Prime" and re-verify jetting performance 
8	**Camera setting**	**Camera Configuration:** 1) Open the **Camera Panel** and select **Enable Camera**. 2) Set the camera to **Index 1** (Index 0 uses the integrated webcam for Zwick laptop). 3) Set the capture interval to **0.50 s** or **1.00 s**.

		4) Define the **Save Directory** on the hardware drive (Organize folders by “*Date_SampleName*”). **NOTE:** Always **re-verify** camera settings when resuming after a printing error. 5) Run calibration   
10	**Start Print**	**Click Print (everything is setting up), and verify stable processing**

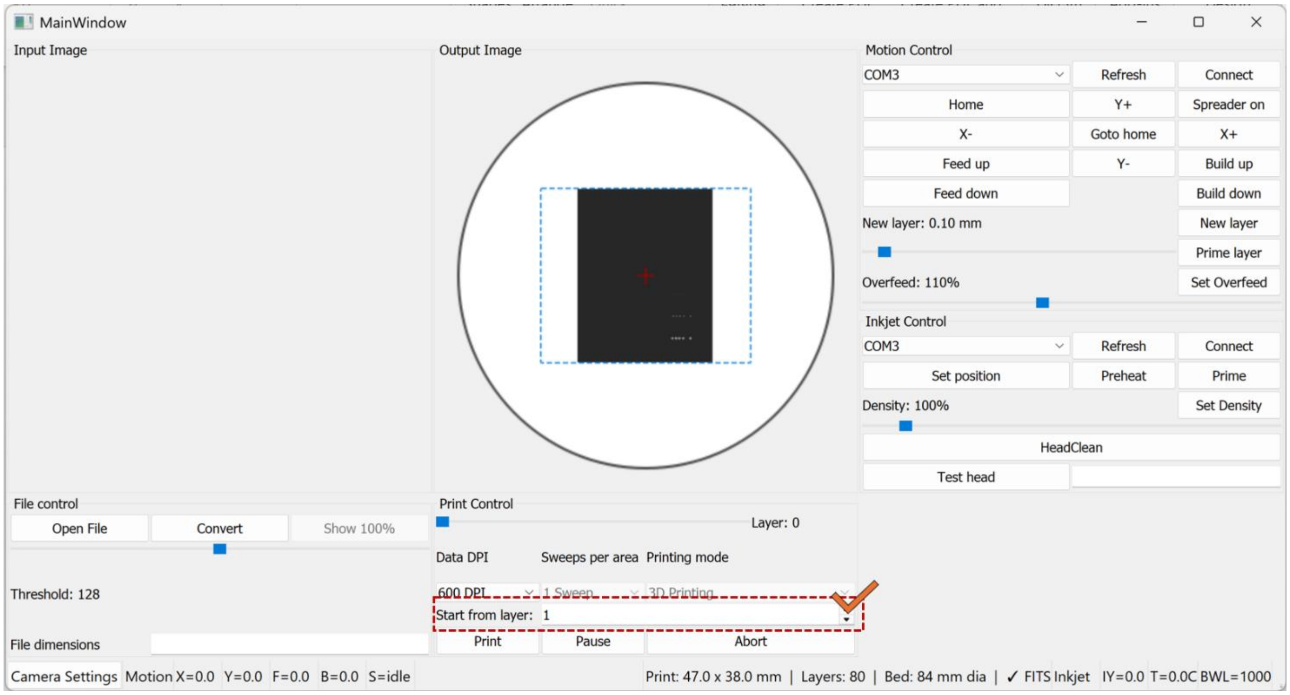
Cleaning Sequence The sequence executes automatically with a **1000 ms interval** between each burst:
Preheat x10 prime x10 Preheat x10 prime x5 preheat x5 → Total 40burst (40seconds)

Important Notes for Cleaning

- Ensure the inkjet is **connected** (Connect button shows "Disconnect").
- Ensure the printer is **not currently printing**.
- The button is **silently blocked** if the inkjet is not connected or if a print is in progress.
- The cleaning sequence runs in a background thread (daemon=True) and does not block the UI.
- Preheat pulse count and Prime pulse count follow the values set in oasis_change_parameters.csv (preheat_pulses, prime_pulses). Defaults: 5000 / 100.
- It is recommended to run HeadClean **before the first print of the day** or after the printhead has been idle for an extended period.

2.4 Start-Layer Control (Mid-Print Resume)

If the printer stopped in the middle of the layer → restart the current layer for the next print run



1. Start-Layer Control (Mid-Print Resume)

Overview A new UI control has been added to allow printing to begin from a specified layer rather than always starting from Layer 1. This is useful when resuming a failed or interrupted print job.

Location in UI Inkjet / File Control panel → **"Start from layer"** spinbox (above the Print button)

How to Use

1. Load an SVG file as normal. The spinbox will activate automatically once the file is loaded.
2. The spinbox maximum is automatically set to the total number of layers in the loaded file.
3. Enter the desired start layer number (minimum: 1).
4. Click **Print** — the printer will begin from the specified layer.

Important Notes

- The spinbox is **disabled** until an SVG file is loaded. It cannot be edited for bitmap files.
- The entered value is automatically clamped: if a value exceeds the total layer count, it is silently reduced to the last valid layer.
- When starting from a middle layer, ensure that the powder bed has already been manually prepared up to the correct height before pressing Print. The software does not verify physical bed state.
- The current_layer_height is initialized from svg_layer_height[start_layer - 1], so Z-height calculations remain accurate regardless of start layer.

3. Print parameter study information

3.1 Parameter Configuration (oasis_change_parameters.csv)

Parameter	Type	Default	Unit	Range	Description
print_speed	variance	2200	mm/min	100–10000	Inkjet sweep speed
dpi	variance	300	dpi	100–1200	Print resolution
layer_passes	variance	3	—	1–5	Repeat passes per layer
density	variance	500	permille	0–1000	Ink firing density
layer_thickness	variance	0.1	mm	0.05–2	Layer thickness
overfeed_percent	variance	100	%	175–500	Default (100) is outside valid range — update to ≥ 175
dwell_time	fixed	5	s	—	Wait time before camera capture

3.2 Experiment Record Management

Each run automatically saves to runs/<RUN_ID>/:

File	Contents
manifest.json	Timestamp, git hash, source CSV path, full resolved config
resolved_params.csv	Original CSV values vs. actually applied values (resolved_value column)
campaign_log.csv	Per-image log: timestamp, step_id, layer_idx, pre_or_post, note, image_filename, and all applied parameter values

To reproduce an experiment: copy the resolved_value column from resolved_params.csv into the value column of oasis_change_parameters.csv and re-run.

3.3 Print Campaign CSV Schema (print_campaign.csv)

Each row defines one step (a group of layers printed under identical parameters). Coworkers edit this file directly to define DOE sequences. Empty cells carry over the previous step's value; step 1 defaults to medium values below.

Column	Code Variable	Low	Medium (default)	High	Notes
step_id	—	—	—	—	Unique integer, sequential
layers	loop count	10	15	20	Layers per step (adjustable per row)
print_speed	self.print_speed	1100	2200	3300	mm/min
travel_speed	self.travel_speed	1800	3000	5000	mm/min

layer_thickness	motion_layer_thickness	0.05 mm	0.1 mm	0.2 mm	mm
dpi	inkjet.SetDPI()	150	300	600	dpi
density	inkjet.SetDensity()	100%	250%	500%	percent
preheat	inkjet.Preheat()	0	1	10	pulse count
prime	inkjet.Prime()	0	15	100	pulse count
layer_passes	layer_passes	1	3	5	passes per layer
overfeed	motion_overfeed	1.75	2.5	5.0	dimensionless multiplier
note	free text label	—	—	—	e.g. “baseline”

3.4 Running a Print Campaign (campaign_runner.py)

campaign_runner.py reads print_campaign.csv and drives the printer autonomously through all steps. Parameters are applied once at the start of each step; existing image capture runs normally every layer. Campaigns can run overnight without intervention.

Pre-campaign checklist

1. Complete standard Pre-Run Checklist (Section 2.1) and GUI Startup Sequence (Section 2.2).
2. Open print_campaign.csv and verify step_id sequence, layers count (10–20), and note labels.
3. Confirm powder reservoir level and image save directory (organized as Date_SampleName).
4. Run a 1-step dry-run (layers=2, note=“smoke_test”) to confirm parameter injection before the full overnight run.

Execution

python campaign_runner.py --csv print_campaign.csv
The runner logs progress to the terminal and writes campaign_log.csv in real time. Image filenames include step_id, layer index, pre/post tag, and note slug (e.g. s001_L003_pre_baseline.png).

Post-campaign

1. Verify campaign_log.csv row count matches expected total images.
2. Review images by step_id to qualitatively confirm defect appearance vs. expected conditions.
3. Copy runs/<RUN_ID>/ and campaign_log.csv to the project archive folder.
4. If mask validation confirms expected defects, proceed to segmentation. Otherwise revise print_campaign.csv and re-run.

Important Notes

- Parameters are applied once per step (at step start), not per layer.
- An empty cell carries over the previous step’s value. Step 1 defaults to medium values.
- A validation warning prints if a value is outside the known range. The print is NOT aborted.
- Do NOT edit print_campaign.csv while the campaign is running.